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Stamp tool, transfer device, and element array manufacturing method

Abstract

A stamp tool whose transport head can be easily shared, a transport device capable of allowing the stamp tool to easily pick up an transport object element disposed on a surface of a substrate from the substrate and transporting the element without being left on the substrate side, and an element array manufacturing method using the same.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a stamp tool, a transport device, and an element array manufacturing method.

BACKGROUND ART

(2) In transport of an extremely small part, use of a stamp-shaped transport tool (stamp tool) having a large number of projections on a surface thereof has been studied. Patent Document 1 below discloses an example of the stamp-shaped transport tool. Conventionally, a stamp tool for allowing detachment of an object to be transported by a coefficient of thermal expansion has been disclosed.

(3) An LED element referred to as a mini-LED or a micro-LED is an example of an extremely small part assumed to be an object to be transported of a stamp tool. The mini-LED or the micro-LED has a width of 1 to 8 μm , a length of 5 to 10 μm , and a height of 0.5 to 3 μm , which are extremely smaller than those of a conventional general LED element.

(4) As in a conventional art, an LED display is manufactured by picking up elements from a wafer on which a plurality of such LED elements is disposed and transporting the elements to a substrate corresponding to a display, and wafers to which LEDs are supplied vary widely depending on the wafer makers and applications.

(5) When transporting the LED elements to the substrate, wafers having different specifications may be exchanged. In this case, it is necessary to use a stamp tool corresponding to each wafer. The stamp tool is replaceably attached to a tip of a general adsorbing mechanism-attached head (transport head). In addition, it is necessary to prepare a plurality of types of transport heads for each of a plurality of types of stamp tools, which complicates the transport device.

(6) Moving to a stamp is performed by bringing adhesive projections formed on the surface of the stamp tool attached to this transport head into contact with an LED on the wafer. At this time, when the stamp tool comes into contact with the surface of the wafer on which the LEDs are fixed, there is a risk that the stamp tool will detach from the transport head due to adhesion of the wafer surface to the stamp tool becoming stronger than an adsorption force of the transport head to the stamp tool.

(7) In particular, when a size of a part, which is an object to be transported, becomes smaller, a size

of the projections of the stamp tool becomes smaller accordingly, and thus a possibility that a flat surface other than the projections will come into contact with a wafer surface increases. When contact of the flat surface of the stamp tool occurs, the contact area increases. Thus, unintended adhesion acts on the stamp tool, which causes a problem in that “the stamp tool adheres to the substrate side such as the wafer and cannot be removed.” [Patent Document 1] US 2017/0173852 A1

SUMMARY

(8) The present invention has been made in view of such circumstances, and an object of the present invention is to provide a stamp tool whose transport head can be easily shared, a transport device capable of allowing the stamp tool to easily pick up an transport object element disposed on a surface of a substrate from the substrate and transporting the element without being left on the substrate side, and an element array manufacturing method using the same.

(9) To achieve the above-mentioned object, a stamp tool according to the present invention includes a stamp layer having a portion allowing an transport object element to detachably adhere thereto, a support plate, the stamp layer being fixed to the support plate, and an adapter plate replaceably attached to the support plate and having a mounting surface, a transport head being allowed to be detachably attached thereon.

(10) With the stamp tool according to the present invention, only the support plate to which the stamp layer is fixed can be replaced from the adapter plate without replacing the entire stamp tool. Therefore, it becomes easier to prepare a stamp tool having different types of stamp layers at low cost. In addition, even when the size of the stamp layer or the size of the support plate is changed, it becomes easy to unify the size of the adapter plate, and it becomes easy to share the transport head or an installation stage. In addition, since the stamp layer is fixed to the support plate, it is easy to ensure a degree of flatness of the stamp surface of the stamp layer.

(11) Preferably, the support plate is replaceably attached to the adapter plate by an adhesive layer. By using the adhesive layer, the support plate can be easily replaceably attached to the adapter plate, and the degree of flatness of the support plate, that is, the degree of flatness of the stamp surface of the stamp layer can be easily ensured.

(12) Preferably, the transport object element has elements formed on a surface of a substrate, projections corresponding to the elements are formed on the stamp layer, and the elements detachably adhere to the respective projections. With such a configuration, it becomes easy to take out a plurality of elements as a plurality of transport object elements from the substrate at the same time and transfer or mount the elements.

(13) Preferably, the support plate has a glass plate or a ceramic plate having a flat surface. With such a configuration, it is easy to ensure the degree of flatness of the support plate, that is, the degree of flatness of the stamp surface of the stamp layer. In addition, in particular, when the support plate is made of a glass plate, it becomes easier to form an adsorbable surface around the stamp layer.

(14) Preferably, a tapered surface having an outer diameter decreasing toward the support plate is formed on a side surface of the adapter plate. A claw portion of a clamping mechanism (also referred to as a chuck mechanism) can be detachably engaged with the tapered surface formed on the side surface of the adapter plate. In addition, a mounting force of the stamp tool to the transport head by the clamping mechanism can be increased. Further, positioning of the stamp tool is facilitated along an inclined surface of a guide member installed in an upper part of the installation stage for the stamp tool.

(15) Preferably, a maximum width of the adapter plate is set to be larger than a width of the support plate. With such a configuration, engagement between the inclined surface of the guide member and the tapered surface of the stamp tool is facilitated.

(16) Preferably, an insertable surface facing the tapered surface of the adapter plate is provided on a surface of the support plate at a side of the adapter plate. The presence of the insertable surface on

the support plate of the stamp tool facilitates detachable engagement of the claw portion of the clamping mechanism with the tapered surface on the side surface of the adapter plate.

(17) Preferably, an adsorbable surface is formed around the stamp layer on a surface of the support plate at a side of the stamp layer. When the adsorbable surface is present on the support plate of the stamp tool, the support plate can be adsorbed on a top surface of the installation stage for the stamp tool, which facilitates sealing and holding the stamp layer inside the housing recess. The stamp layer in the housing recess is kept clean.

(18) A shim plate for adjusting a degree of parallelism (a degree of flatness) of the support plate may be interposed between the stamp layer and the adapter plate. With such a configuration, the degree of flatness of the support plate is improved, and the degree of flatness of the stamp surface is improved.

(19) A method of manufacturing an element array according to the present invention includes taking out transport object elements from a substrate simultaneously and transporting the elements using any one of the above-described the stamp tools. In the method of manufacturing an element array according to the present invention, the element array having the plurality of elements can be easily manufactured.

(20) In addition, to achieve the above-mentioned object, a transport device according to the present invention is a transport device including a stamp tool, and a transport head configured to detachably transport the stamp tool, in which the stamp tool includes a stamp layer allowing an transport object element disposed on a substrate with a predetermined fixing force $F1$ to adhere into the stamp layer with predetermined adhesion force $F2$, and an adapter plate configured to support the stamp layer, having a mounting surface allowing the transport head to be detachably attached thereon, a mounting force $F3$ to the mounting surface of the adapter plate by the transport head is greater than the fixing force $F1$, and the adhesion force $F2$ of the stamp layer to the transport object element is greater than the fixing force $F1$.

(21) In the transport device according to the present invention, the mounting force $F3$ to the mounting surface of the adapter plate by the transport head is greater than the fixing force $F1$, and the adhesion force $F2$ of the stamp layer to the transport object element is greater than the fixing force $F1$. Therefore, the stamp tool can easily pick up and transport the transport object element disposed on the surface of the substrate from the substrate without being left on the substrate side.

(22) Preferably, the mounting force $F3$ to the mounting surface of the adapter plate by the transport head includes a primary mounting force $F3a$ by a primary attaching means and a secondary mounting force $F3b$ by a secondary attaching means other than the primary attaching means.

Preferably, the primary attaching means includes a vacuum suction hole formed in the transport head. By only providing the secondary attaching means to a general transport head having the vacuum suction hole, the mounting force $F3$ to the mounting surface of the adapter plate by the transport head is easily set to be larger than the fixing force $F1$ of the transport object element with respect to the substrate.

(23) Preferably, the secondary attaching means includes at least one of a clamping mechanism (also referred to as a chuck mechanism) for detachably attaching the adapter plate to the transport head, an electrostatic adsorption mechanism, a fitting mechanism, and a screwing mechanism. By providing these mechanisms to the transport head, it becomes easy to set the mounting force $F3$ to the mounting surface of the adapter plate by the transport head to be greater than the fixing force $F1$ of the transport object element with respect to the substrate.

(24) Preferably, a tapered surface having an outer diameter decreasing toward the stamp layer is formed on a side surface of the adapter plate. In addition, preferably, a claw portion of a clamping mechanism included in the secondary attaching means is allowed to be engaged with the tapered surface formed on the side surface of the adapter plate. With such a configuration, the claw portion of the clamping mechanism can easily be detachably engaged with the tapered surface on the side surface of the adapter plate. In addition, the mounting force $F3$ of the stamp tool with respect to the

transport head by the clamping mechanism can be increased.

(25) The stamp tool further includes a support plate on which the stamp layer is fixed and to which the adapter plate is replaceably attached to the support plate. With such a configuration, only the support plate to which the stamp layer is fixed can be replaced from the adapter plate without replacing the entire stamp tool. Therefore, it becomes easier to prepare a stamp tool having different types of stamp layers at low cost.

(26) Preferably, the support plate is replaceably attached to the adapter plate by an adhesive layer. The adhesion by this adhesive layer is preferably larger than the fixing force F_1 of the transport object element with respect to the substrate.

(27) The transport object element may have elements formed on a surface of a substrate, and preferably, a plurality of projections corresponding to the elements is formed on the stamp layer, and the elements detachably adhere to the respective projections. With such a configuration, the plurality of elements as the plurality of transport object elements can be taken out from the substrate at the same time.

(28) A method of manufacturing an element array according to the present invention includes a step of simultaneously taking out and transporting a plurality of elements as a plurality of transport object elements from the substrate simultaneously and transporting the elements using any one of the above-described transport devices. The method of manufacturing the element array according to the present invention can easily manufacture an element array having a plurality of elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic front view and a main part enlarged view of a stamp tool used in a transport device according to an embodiment of the present invention;

(2) FIG. 1B is a schematic front view of a stamp tool according to another embodiment of the present invention;

(3) FIG. 1C is a schematic plan view of a modification example of the stamp tool of FIG. 1B;

(4) FIG. 1D is a cross-sectional view of the stamp tool taken along ID-ID illustrated in FIG. 1C;

(5) FIG. 1E is a cross-sectional view of the stamp tool taken along IE-IE illustrated in FIG. 1C;

(6) FIG. 2A is a schematic diagram of a transport device including a transport head that detachably transports the stamp tool illustrated in FIG. 1A;

(7) FIG. 2B is a schematic diagram of a transport device illustrating a state in which the stamp tool is gripped by the transport head illustrated in FIG. 2A;

(8) FIG. 3A is a schematic diagram of a transport device illustrating a state before picking up an element from a semiconductor substrate;

(9) FIG. 3B is a schematic diagram of the transport device illustrating a state of pressing a stamp layer of the stamp tool against the element on the semiconductor substrate from the state illustrated in FIG. 3A;

(10) FIG. 3C is a schematic diagram of the transport device illustrating a state after picking up the element from the semiconductor substrate;

(11) FIG. 4A is a partial schematic diagram illustrating details of a claw portion of a clamping mechanism used in a transport device according to another embodiment of the present invention;

(12) FIG. 4B is a partial schematic diagram illustrating details of a claw portion of the clamping mechanism used in the transport device according to the other embodiment of the present invention;

(13) FIG. 5A is a schematic cross-sectional view of the element formed on the semiconductor substrate;

(14) FIG. 5B is a schematic cross-sectional view illustrating a state in which the element on the

semiconductor substrate is picked up by the stamp tool of the transport device;

(15) FIG. 5C is a schematic cross-sectional view illustrating a state in which the element on the semiconductor substrate is picked up by the stamp tool of the transport device and then disposed on a first transfer substrate (sheet);

(16) FIG. 5D is a schematic cross-sectional view illustrating a state in which an element array disposed on the first transfer substrate (sheet) is transferred to a second transfer substrate (sheet);

(17) FIG. 5E is a schematic cross-sectional view illustrating a state before the element array disposed on the second transfer substrate (sheet) is transferred to a mounting substrate (sheet);

(18) FIG. 5F is a schematic cross-sectional view illustrating a state after the element array disposed on the second transfer substrate (sheet) is transferred to the mounting substrate (sheet);

(19) FIG. 6 is a schematic perspective view of a stamp stage on which the stamp tool illustrated in FIG. 1A is installed; and

(20) FIG. 7 is a schematic cross-sectional view of the stamp stage taken along line VII-VII illustrated in FIG. 6.

DETAILED DESCRIPTION

(21) Hereinafter, the present invention will be described based on embodiments illustrated in the drawings.

First Embodiment

(22) As illustrated in FIG. 2A, a transport device **20** according to the present embodiment includes a stamp tool **10** and a transport head **22**. As illustrated in FIG. 1A, the stamp tool **10** includes a stamp layer **12**, a support plate **14**, and an adapter plate **16**.

(23) On the stamp layer **12**, projections **11** protruding downward along a Z-axis are formed in a matrix at predetermined intervals in an X-axis direction and a Y-axis direction. For example, an X-axis direction width $x1$ of the projections **11** and an X-axis direction interval $x2$ of adjacent projections **11** are determined according to an X-axis direction width $x3$, an X-axis direction interval $x4$, etc. of elements (an example of an transport object element) **32r** for red light emission mounted on a surface of a mounting substrate (the substrate may be a sheet, which is similarly applied hereinafter) illustrated in FIG. 5F.

(24) Note that although not illustrated in FIG. 1A, the above description is similarly applied to a Y-axis direction width of the projections **11** and a Y-axis direction interval of adjacent projections **11**. The projections **11** are disposed in a matrix on a lower surface of the stamp layer **12**, and the number of projections disposed is not particularly limited, and is one to several hundred thousand.

(25) In the present embodiment, in the drawings, an X-axis (first axis), a Y-axis (second axis), and the Z-axis (third axis) are substantially perpendicular to one another, the X-axis and Y-axis are parallel to a planar direction of the stamp layer **12**, and the Z-axis is parallel to a direction in which the projections **11** protrude.

(26) As illustrated in FIG. 1A, a protrusion height $z1$ of the projections **11** of the stamp layer **12** is determined in relation to a Z-axis direction height $z2$ of the elements **32r** illustrated in FIG. 5B, and is preferably, for example, 1 to 8 times the Z-axis direction height $z2$. A Z-axis direction thickness $z3$ of the stamp layer **12** is not particularly limited, and is preferably about 0.25 times or more the protrusion height $z1$ of the projections **11**. Note that the X-axis direction width $x3$ (the Y-axis direction width is about the same) of the elements **32r** is, for example, 1 to 150 μm , and the height $z2$ is, for example, 1 to 150 μm .

(27) The stamp layer **12** and the projections **11** may be made of different materials as long as the stamp layer **12** and the projections **11** are strongly bonded, or may be made of the same material. By using the same material, a possibility that the projections **11** will be peeled from the stamp layer **12** is reduced. At least the projections **11** are made of an adhesive material, and are configured to allow the elements **32r** disposed with a predetermined fixing force $F1$ on an element forming substrate **30** illustrated in FIG. 5B to adhere thereto with predetermined adhesion force $F2$. The material and shape of the projections **11** are determined so that, when a lower end of a projection **11**

is pressed against an upper surface of an element **32r** with a predetermined force, the adhesion force **F2** of the projection **11** with respect to the element **32r** becomes greater than the fixing force **F1** of the element **32r** with respect to the substrate **30**.

(28) The material of the projections **11** is not particularly limited, and examples thereof include polydimethylsiloxane (PDMS), organosilicon compounds, and viscoelastic elastomer such as polyether rubber. The stamp layer **12** may be made of the same material as that of the projections **11**, and a surface of the stamp layer **12** other than the projections **11** is preferably non-adhesive. It is preferable not to pick up the elements **32r** by adhesion except for the projections **11**.

(29) As illustrated in FIG. **1A**, the stamp layer **12** is fixed to support plate **14**. The support plate **14** is made of a material having higher rigidity and a superior degree of flatness than the stamp layer **12**, and is preferably made of a glass plate, a metal plate, a ceramic plate, etc. A thickness of the support plate **14** is not particularly limited, and is preferably 0.5 mm or more.

(30) The stamp layer **12** may be formed directly on a surface of the support plate **14**, or may be fixed by an adhesive layer. In any case, the stamp layer **12** is fixed to the surface of the support plate **14** with a sticking force sufficiently higher than the adhesion force **F2** illustrated in FIG. **5B**. In a post-process, the element **32r** is peeled off from the projection **11** and disposed on, for example, a first transfer substrate (sheet) **50** illustrated in FIG. **5C**. Thus, at that time, it is important that the stamp layer **12** does not peel off from the support plate **14**.

(31) As illustrated in FIG. **1A**, the support plate **14** is detachably fixed to an adhesive surface **16b** of the adapter plate **16** by an adhesive layer **15** on a surface opposite to the stamp layer **12**. Adhesion between the support plate **14** and the adapter plate **16** by the adhesive layer **15** is adhesion sufficiently higher than the adhesion force **F2** illustrated in FIG. **5B**. However, the support plate **14** can be removed from the adhesive surface **16b** of the adapter plate **16** when replacing the stamp layer **12** after repeated use. The adhesive layer **15** may include a double-sided adhesive tape, etc.

(32) An X-axis direction width and a Y-axis direction width of the support plate **12** are preferably larger than those of the stamp layer **12**, and larger than an X-axis direction width and a Y-axis direction width of the adhesive surface **16b** of the adapter plate **16**. On a surface of the support plate **14** on the stamp layer side, a flat adsorbable surface **14b**, on which the stamp layer **12** is not formed, is formed around the stamp layer **12**. In the present embodiment, the stamp layer **12** has a rectangular shape when viewed in the Z-axis direction. However, the support plate **14** may have a rectangular or circular shape. The adsorbable surface **14b** can be detachably attached to a top surface of an installation stage **82** illustrated in FIG. **7**.

(33) An upper surface of the adapter plate **16** opposite to the adhesive surface **16b** is a flat mounting surface **16a**. At least both side surfaces of the adapter plate **16** in the X-axis direction are tapered surfaces **16c** so that the area of the mounting surface **16a** is larger than the area of the adhesive surface **16b**. That is, the tapered surface **16c**, outer diameters of which decrease toward the stamp layer **12**, are formed on at least the side surfaces of the adapter plate **16** in the X-axis direction. The tapered surfaces **16c** may be formed on both side surfaces of the adapter plate **16** in the Y-axis direction, or the tapered surfaces **16c** may be formed along the entire circumference of the side surfaces of the adapter plate **16**.

(34) In the present embodiment, the tapered surfaces **16c** are also formed on the both side surfaces of the adapter plate **16** in the Y-axis direction, and the tapered surfaces **16c** are formed along the entire circumference of the side surfaces of the adapter plate **16**. In the present embodiment, the adapter plate **16** has a rectangular shape when viewed in the Z-axis direction, and at least a maximum X-axis direction width of the adapter plate **16** is preferably larger than the X-axis direction width of the support plate **14**. Note that as illustrated in FIG. **7**, a maximum Y-axis direction width of the adapter plate **16** may be substantially equal to the Y-axis direction width of the support plate **14**, or may be larger or smaller than the Y-axis direction width of the support plate **14**.

(35) On a surface of the support plate **14** opposite to the adsorbable surface **14b** illustrated in FIG. **1A**, a flat insertable surface **14c** facing the tapered surfaces **16c** is formed around the adhesive surface **16b** of the adapter plate **16**. On the insertable surface **14c** located on both sides in the X-axis direction, a claw portion **26a** of a chuck mechanism **26** illustrated in FIG. **2B** is engaged with each of the tapered surfaces **16c** of the adapter plate **16**. In addition, an inclined surface **89** of a guide member **88** of an installation stage **82** illustrated in FIG. **6** is engaged with each of the tapered surfaces **16c** of the adapter plate **16** located on both sides in the X-axis direction.

(36) A thickness of the adapter plate **16** illustrated in FIG. **1A** in the Z-axis direction is sufficiently larger than the thickness of the support plate **14**, and is preferably 1.2 times or more, and preferably about 2 to 6 times the thickness of the support plate **14**. Note that an edge portion **16d** including a chamfered portion or an R-portion is formed on an outer peripheral edge portion of the mounting surface **16a** on the upper surface of the adapter plate **16**. Note that it is preferable that a chamfered portion or an—R portion is formed on the outer peripheral edge portion of the mounting surface **16a** on the upper surface of the adapter plate **16**.

(37) Tip surfaces **92** of a pair of positioning members **90** illustrated in FIGS. **6** and **7** come into contact with edge portions **16d** of the adapter plate **16** located on both sides in the Y-axis direction to position a Y-axis direction position of the stamp tool **10** placed on the installation stage **82**. Positioning of an X-axis direction position of the stamp tool **10** is performed by the inclined surface **89** of the guide member **88** of illustrated in FIG. **6** and a claw portion **26a** of a clamping mechanism **26** of the transport device **20** illustrated in FIG. **2B**.

(38) An adsorbing surface **24** of the transport head **22** of the transport device **20** illustrated in FIG. **2A** can be adsorbed to the mounting surface **16a** on the upper surface of the adapter plate **16**. A vacuum suction hole serving as a primary attaching means is formed on the adsorbing surface **24** of the transport head **22**, and by generating a negative pressure in the vacuum suction hole, the mounting surface **16a** of the adapter plate **16** of the stamp tool **10** is vacuum-adsorbed to the adsorbing surface **24**. A vacuum adsorption force of the adsorbing surface **24** to the mounting surface **16a** of the adapter plate **16** of the stamp tool **10** is assumed to be a primary mounting force **F3a**.

(39) In addition, in the present embodiment, the chuck mechanism **26** is attached on the transport head **22** via an opening/closing mechanism **28**. The claw portion **26a** is formed inside the chuck mechanism **26**. The chuck mechanism **26** including the claw portion **26a** is moved, for example, in the X-axis direction by the opening/closing mechanism **28**, so that the claw portion **26a** opens the entire lower surface of the adsorbing surface **24** as illustrated in FIG. **2A**, or the claw portion **26a** is located below each of both sides of the adsorbing surface **24** in the X-axis direction as illustrated in FIG. **2B**.

(40) A tapered engaging surface **26b** is formed on each claw portion **26a**. The tapered surface of the engaging surface **26b** is adapted to a shape of the tapered surface **16c** of the adapter plate **16** of the stamp tool **10**, and can be engaged with the tapered surface **16c**. As illustrated in FIGS. **2A** and **2B**, before the mounting surface **16a** of the adapter plate **16** is adsorbed to the adsorbing surface **24** of the transport head **22**, the claw portion **26a** of the chuck mechanism **26** is opened by the opening/closing mechanism **28**. After the mounting surface **16a** of the adapter plate **16** is adsorbed to the adsorbing surface **24** of the transport head **22**, the chuck mechanism **26** moves in a direction in which the claw portion **26a** is closed by the opening/closing mechanism **28**, and the engaging surface **26b** is engaged with the tapered surface **16c**.

(41) As a result, the stamp tool **10** is attached on the transport head **22** with a total mounting force **F3** of a primary mounting force **F3a** generated by a vacuum suction hole serving as a primary attaching means formed in the transport head **22** and a secondary mounting force **F3b** generated by the chuck mechanism **26** as serving as a secondary attaching means. As the transport head **22** becomes smaller, the primary mounting force **F3a** alone generated by the vacuum suction hole of the transport head **22** tends to hardly become larger than the fixing force **F1** illustrated in FIG. **5B**.

In the present embodiment, when the secondary mounting force $F3b$ generated by the chuck mechanism **26** serving as the secondary attaching means is added to the primary mounting force $F3a$, the total mounting force $F3$ ($=F3a+F3b$) reliably becomes larger than the fixing force $F1$.

(42) Next, a description will be given of a method for manufacturing a display element array using the transport device **20** having the stamp tool **10** according to present embodiment.

(43) First, the transport device **20** illustrated in FIG. 2A picks up the stamp tool **10** disposed on the installation stage **82** illustrated in FIG. 6. In the present embodiment, at least three stamp tools may be prepared, for example for the three primary colors of light R, G and B, and each stamp tool is preferably installed on each installation stage **82**. Alternatively, the installation stage **82** is replaced with respect to a base **80** for each of the stamp tools **10** for R, G and B.

(44) In the following description, one installation stage **82** will be described. As illustrated in FIGS. 6 and 7, the block-shaped installation stage **82** is installed on the base **80** in a detachable and replaceable manner using bolts, etc. As illustrated in FIG. 7, a housing recess **86** and a top surface **84** surrounding the housing recess **86** are formed in an upper part of the installation stage **82** in the Z-axis direction. The housing recess **86** is formed, for example, by counterbore-molding a center of an upper surface of the square pole-shaped stage **82**. As illustrated in FIG. 7, the housing recess **86** is adapted to completely accommodate the stamp layer **12** of the stamp tool **10**.

(45) In addition, suction holes **85** are formed at a plurality of locations in a circumferential direction on the top surface **84** formed around the housing recess **86** to detachably adsorb and hold the adsorbable surface **14b** of the support plate **14** on the top surface **84**. In addition, a plurality of gas flow holes **83** formed in the stage **82** communicates with the housing recess **86**. By adsorbing the adsorbable surface **14b** of the support plate **14** on the top surface **84**, the housing recess **86** can be sealed except for the gas flow holes **83**. By allowing cleaning gas to flow into a housing space **86** through the gas flow holes **83**, dust and impurities adhering to the stamp layer **12** can be discharged to the outside.

(46) Two guide members **88** are detachably attached by bolts, etc. at each side of the stage **82** on both side surfaces substantially perpendicular to the X-axis. The inclined surface **89** is formed on an upper side of an inner surface of the guide member **88**. The tapered surface **16c** of the adapter plate **16** illustrated in FIG. 1A can be in contact with each inclined surface **89**, and the tapered surface **16c** of the adapter plate **16** facing in the X-axis direction slides along each inclined surface **89**. Therefore, the adapter plate **16** of the stamp tool **10** is dropped onto the stage **82** while sliding on the inclined surface **89**, and the stamp layer **12** is housed inside the housing recess **86** as illustrated in FIG. 7. In addition, rough positioning of the stamp tool **10** with respect to the stage **82** in the X-axis direction is performed.

(47) As illustrated in FIG. 6, four guide members **88** are attached to the stage **82** so as to be located inside both edge portions **16d** of the adapter plate **16** of the stamp tool **10** in the Y-axis direction. The positioning members **90** in the Y-axis direction are disposed on both sides of the stage **82** in the Y-axis direction so as to be movable in the Y-axis direction. The tip surfaces **92** are formed on the positioning members **90**, respectively, these tip surfaces **92** face each other along the Y-axis, and each of the tip surfaces **92** can come into contact with the edge portion **16d** of the adapter plate **16** in the Y-axis direction as illustrated in FIG. 7. The stamp tool **10** is positioned with respect to the stage **82** in the Y-axis direction by the tip surface **92** coming into contact with the edge portion **16d** of the adapter plate **16** in the Y-axis direction.

(48) Next, a description will be given of a method of picking up the stamp tool **10** from the installation stage **82** illustrated in FIGS. 6 and 7 using the transport device **20** illustrated in FIG. 2A.

(49) First, the positioning members **90** are used to position the stamp tool **10** on the stage **82** in the Y-axis direction. Thereafter, the stage **82** moves together with the base **80** and is positioned below the transport head **22** of the transport device **20** illustrated in FIG. 2A. Note that the transport head **22** may be moved without moving the stage **82**, or both of the transport head **22** and the stage **82**

may be moved. The transport head **22** may be rotated around the Z-axis as necessary.

(50) After the stamp tool **10** on the stage **82** is positioned under the transport head **22** in the Z-axis, the head **22** is moved downward along the Z-axis so that a lower end of the transport head **22** is brought into contact with the mounting surface **16a** of the adapter plate **16**, and vacuum adsorption by the transport head **22** is started. Next, as illustrated from FIG. 2A to FIG. 2B, the clamping mechanism **26** is closed, and the engaging surfaces **26b** of the claw portions **26a** are engaged with the tapered surfaces **16c** located on both sides of the adapter plate **16** in the X-axis direction, respectively.

(51) Thereafter, the pair of positioning members **90** illustrated in FIG. 7 are opened in the Y-axis direction to release contact between the edge portion **16d** of the adapter plate **16** and the tip surface **92**. Before and after the release, adsorption of the support plate **14** to the top surface **84** of the stage by the suction holes **85** of the stage **82** is released. Thereafter, when the transport head **22** is moved upward along the Z-axis, as illustrated in FIG. 2B, the stamp tool **10** is positioned at a lower end of the transport head **22** in the X-axis and Y-axis and is held in a state where levelness of the stamp tool **10** is maintained.

(52) In a state where the stamp tool **10** is attached on the transport head **22** as illustrated in FIG. 2B, the transport device **20** is relatively moved in the X-axis and Y-axis directions, and is located above the element **32r** formed on a surface of the element forming substrate **30** installed on an installation table **40** as illustrated in FIG. 3A. As illustrated in FIG. 5A, on the surface of the element forming substrate **30**, for example, an element **32r** for red light emission, an element **32g** for green light emission, or an element **32b** for blue light emission is incorporated. The substrate **30** may be, for example, a sapphire substrate, a glass substrate, a GaAs substrate, a SiC substrate, etc., depending on the type of device (blue light emitting device, red light emitting device, green light emitting device, etc.).

(53) In the present embodiment, the elements **32r**, **32g**, and **32b** are, for example, micro LED elements. Note that in the following description, only the element **32r** will be described. However, the other elements **32g** and **32b** are also processed in a similar manner using separate stamp tools **10**, respectively. The stamp tool **10** is preferably prepared for each type of the different elements **32r**, **32g**, and **32b**. However, the transport head **22** may be used in common. The stamp tool **10** in a standby state is installed, for example, on the stage **82** illustrated in FIGS. 6 and 7, and the stamp layer **12** is sealed inside the housing recess **86** and kept clean.

(54) As illustrated from FIG. 3A to FIG. 3B, the transport device **20** is moved downward in the Z-axis direction to press the projection **11** of the stamp tool **10** against an upper surface of the element **32r** of the substrate **30**. As a result, the element **32r** adheres to the projection **11**. Thereafter, as illustrated in FIG. 3C, the stamp tool **10** is lifted upward in the Z-axis direction together with the transport device **20**. As a result, as illustrated in FIG. 5B, the element **32r** adheres to each projection **11**, and the element **32r** is picked up from the substrate **30** together with the projection **11**. The element **32r** left on the substrate **30** is similarly picked up by the stamp tool **10** of the transport device **20** later.

(55) Next, the element **32r** picked up by the projection **11** of the stamp tool **10** is transported by the transport device **20** onto the first transfer substrate (the substrate may be a sheet/which is similarly applied hereinafter) **50** illustrated in FIG. 5C and disposed on an adhesion layer **52**. Alternatively, the element **32r** picked up by the projection **11** of the stamp tool **10** may be transported, for example, by the transport device **20** onto a mounting substrate **70** illustrated in FIG. 5F and disposed on the mounting substrate **70** there. In the present embodiment, a transfer method is used for description.

(56) An array of the elements **32r** adhering to the projections **11** of the stamp layer **12** illustrated in FIG. 5B is transferred onto the adhesion layer **52** of the substrate **50** including an adhesion sheet, etc. illustrated in FIG. 5C. To this end, the elements **32r** adhering to the projections **11** of the stamp layer **12** are pressed against the surface of the adhesion layer **52**, and then the stamp layer **12** is

lifted together with the transport device **20**. As a result, a plurality of elements **32r** is transferred to the surface of the adhesion layer **52** at the same time. Note that before the transfer, the transport device **20** illustrated in FIG. **3C** is moved onto the substrate **50** illustrated in FIG. **5C** by a transport mechanism of the transport device **20**.

(57) The adhesion of the adhesion layer **52** of the adhesive sheet including the substrate **50** is adjusted so that the adhesion of the adhesion layer **52** is greater than the adhesion of the projections **11**. The adhesion layer **52** is made of, for example, an adhesion resin such as natural rubber, synthetic rubber, acrylic resin, or silicone rubber, and a thickness z_4 thereof is preferably about 0.5 to 2.0 times the height z_2 (see FIG. **5B**) of the elements **32r**. Note that in order to smoothly move the element **32r** from the projection **11** to the adhesion layer **52**, an operation (for example, applying heat) may be added to facilitate peeling of the element **32r** from the projection **11**.

(58) The other elements **32g** and **32b** are also transferred to the adhesion layer **52** of the substrate **50** in a similar manner as described above. Three elements **32r**, **32g**, and **32b** of R, G, and B constitute one pixel unit, and by disposing these pixel units in a matrix, a color display screen can be obtained.

(59) The other elements **32g** and **32b** are also transferred to the adhesion layer **52** of the substrate **50** in a similar manner as described above. Three elements **32r**, **32g**, and **32b** of R, G, and B constitute one pixel unit, and by disposing these pixel units in a matrix, a color display screen can be obtained.

(60) Next, as illustrated in FIG. **5D**, the entire array of the three elements **32r**, **32g**, and **32b** disposed on the surface of the first transfer substrate **50** is transferred to the adhesion layer **62** of the second transfer substrate **60**, so that respective terminals of the elements **32r**, **32g**, and **32b** are disposed to face the outside of the substrate **60**. For this transfer, a technique such as a laser lift method may be used, and a transfer method using a difference in adhesion, a transfer method involving heat peeling, etc. may be used. With the terminals of the elements **32r**, **32g**, and **32b** facing the outside of the substrate **60**, a tin plating film may be formed on each terminal using an electroless plating method, etc.

(61) Next, as illustrated in FIGS. **5E** and **5F**, the entire array of the three elements **32r**, **32g**, and **32b** is transferred from the adhesion layer **62** of the substrate **60** to the mounting substrate **70**. For the transfer, a technique such as the laser lift method may be used, and a transfer method using a difference in adhesion, a transfer method involving heat peeling, etc. may be used.

(62) Note that after the transfer, in order to connect the terminal of each of the elements **32r**, **32g**, and **32b** to a circuit pattern of the mounting substrate, for example, it is preferable that anisotropic conductive paste (ACP) is applied to the surface of the mounting substrate **70** or an anisotropic conductive film (ACF) is disposed thereon. As illustrated in FIG. **5F**, after disposing the elements **32r**, **32g**, and **32b** on the substrate **70** via the ACP or ACF, the respective elements **32r**, **32g**, and **32b** may be pressed toward the substrate **70** and heated using a heating/pressurizing device (not illustrated). As a result, the terminals of each of the elements **32r**, **32g**, and **32b** can be connected to the circuit pattern of the mounting substrate.

(63) In the transport device **20** according to the present embodiment, the mounting force F_3 to the mounting surface **16a** of the adapter plate **16** by the transport head **22** illustrated in FIG. **2B** is greater than the fixing force F_1 illustrated in FIG. **5B**, and the adhesion force F_2 of the projection **11** of the stamp layer **12** on the element **32r** is greater than the fixing force F_1 . Therefore, the stamp tool **10** can easily pick up the element **32r** disposed on the surface of the substrate **30** from the substrate **30** and transport the element **32r** without being left on the substrate **30** side.

(64) In addition, in the present embodiment, the mounting force F_3 to the mounting surface **16a** of the adapter plate **16** by the transport head **22** illustrated in FIG. **2B** is a sum of the primary mounting force F_{3a} corresponding to the adsorption force of the vacuum suction hole and the secondary mounting force F_{3b} generated by the clamping mechanism **26** serving as the secondary attaching means. That is, in the present embodiment, only by providing the clamping mechanism

26 to a general transport head 22 having a vacuum suction hole, it becomes easy to set the mounting force F3 to the mounting surface 16a of the adapter plate 16 by the transport head 22 to be greater than the fixing force F1 of the element 32r to the substrate 30 illustrated in FIG. 5B.

(65) Further, in the present embodiment, on both side surfaces of the adapter plate 16 in the X-axis direction, the tapered surfaces 16c are formed to decrease in outer diameter toward the stamp layer 12. In addition, the claw portion 26a of the clamping mechanism 26 can be engaged with the tapered surface 16c. With this configuration, the claw portion 26a of the clamping mechanism 26 can be easily engaged with the tapered surface 16c on the side surface of the adapter plate 16 in a detachable manner. In addition, the mounting force F3 of the stamp tool 10 to the transport head 22 by the clamping mechanism 26 can be increased.

(66) In addition, when the tapered surfaces 16c are formed to decrease in outer diameter toward the stamp layer 12 on both side surfaces of the adapter plate 16 in the X-axis direction, rough positioning of the stamp tool 10 in the X-axis direction is facilitated along the inclined surface 89 of the guide member 88 installed in an upper part of the stage 82 illustrated in FIG. 6. In particular, as illustrated in FIG. 1A, when the maximum width of the adapter plate 16 in the X-axis direction is larger than the width of the support plate 14, the inclined surface 89 of the guide member 88 and the tapered surface 16c of the stamp tool 10 are easily engaged with each other.

(67) In addition, the presence of the insertable surface 14c on the support plate 14 of the stamp tool 10 facilitates detachable engagement of the claw portion 26a of the clamping mechanism 26 with the tapered surface 16c on the side surface of the adapter plate 16. Further, when the adsorbable surface 14b is present on the support plate 14 of the stamp tool 10, as illustrated in FIG. 7, the support plate 14 can be adsorbed on the top surface 84 of the stage 82, and the stamp layer 12 is easily sealed and held inside the housing recess 86. The adsorbable surface 14b can be easily formed around the stamp layer 12 by configuring the support plate 14 using a glass plate, etc.

(68) The stamp tool 10 further includes the support plate 14 to which the stamp layer 12 is fixed and to which an adapter plate 16 is replaceably attached. With this configuration, only the support plate 14 to which the stamp layer 12 is fixed can be replaced from the adapter plate 16 without replacing the entire stamp tool 10. Therefore, it becomes easy to prepare the stamp tool 10 having different types of stamp layers 12 at low cost. In addition, by using the adapter plate 16 in common, it is unnecessary to use different types of transport heads in accordance with the stamp tools, and the overall structure of the transport device can be simplified.

(69) In the present embodiment, the plurality of projections 11 corresponding to the elements 32r (32g, 32b) is formed on the stamp layer 12, and the elements 32r (32g, 32b) detachably adhere to the respective projections 11. With this configuration, a plurality of elements 32r (32g, 32b) may be taken out from the substrate 30 at the same time. In the element array manufacturing method of the present embodiment, an element array having the plurality of elements 32r (32g, 32b) can be easily manufactured.

(70) In addition, in the present embodiment, as illustrated in FIG. 6, the installation stage 82 is replaceably attached to the base 80. Therefore, the stage 82 corresponding to the stamp tool 10 may be prepared, and only the stage 82 may be replaced when changing to a different type of stamp tool 10. Since a degree of flatness of the stage 82 is ensured with respect to the base 80, there is no need to adjust a degree of flatness of the stamp tool 10 when the stamp tool 10 is replaced.

(71) Therefore, in the present embodiment, the transport head 22 can favorably pick up the stamp tool 10 from the installation stage 82 without causing an error in adsorption by the transport head 22 or an error in gripping by the clamping mechanism 26.

Second Embodiment

(72) As illustrated in FIG. 1B, in a stamp tool 10a used in a transport device of the present embodiment, a shim plate 18 is interposed between the stamp layer 12 and the adapter plate 16 to adjust a degree of parallelism of the support plate 14. An inclined surface 14a is formed on a part of a side surface of the support plate, and the shim plate 18 is engaged with the inclined surface 14a

so that the degree of parallelism of the support plate **14** can be adjusted. The degree of parallelism of the support plate **14** can be adjusted using a mode in which the shim plate **18** is installed between the support plate **14** and the adapter plate **16** via the adhesive layer **15**.

(73) Note that since the purpose of installing the shim plate **18** is to adjust a degree of parallelism, a position where the shim plate **18** is installed is not limited thereto. The shim plate **18** may be installed across the entire periphery of the adapter plate **16**, or may be installed intermittently. For example, as illustrated in FIGS. **1C**, **1D**, and **1E**, adhesive layers **15** may be provided at four corners of the adapter plate **16**, and shim plates **18** may be provided only at two locations on one side in the Y-axis direction between the adhesive surface **16b** and the support plate **14** with the adhesive layer **15** interposed therebetween. With this configuration, when the adapter plate **16** (or the support plate **14**) has a rectangular shape, a small amount of degree of parallelism can be adjusted.

(74) That is, as illustrated in FIG. **1C**, when the adapter plate **16** (or the support plate **14**) has a rectangular shape, the degree of parallelism can be adjusted by disposing the shim plate **18** on one of opposing sides. In addition, when the adapter plate **16** (or the support plate **14**) has a circular shape, the degree of parallelism can be adjusted by disposing the shim plate **18** in one of circular arc regions at point-symmetrical positions.

(75) More specifically, for example, as illustrated in FIG. **1E**, when a thickness of the stamp layer **12** varies along the Y-axis direction, the shim plate **18** may be disposed in a gap between the support plate **14** and the adapter plate **16** on one side in the Y-axis direction. In this way, the mounting surface **16a** and a stamp surface of the stamp layer **12** become parallel to each other, and the degree of parallelism can be adjusted. Note that, in FIG. **1E**, in order to facilitate description, thicknesses and inclinations of the stamp layer **12**, the shim plate **18**, and the adhesive layer **15** are illustrated to be larger than actual thicknesses and inclinations.

(76) Other configurations and effects of the transport device and the stamp tool of the present embodiment are the same as those of the first embodiment, and a detailed description thereof will be omitted.

Third Embodiment

(77) As illustrated in FIG. **4A**, in the transport device of the present embodiment, an elastically deformable engaging projection **26c** is attached on the engaging surface **26b** of the claw portion **26a** of the chuck mechanism **26**, and the engaging projection **26c** can be engaged with the tapered surface **16c** of the adapter plate **16**. The engaging projection **26c** is made of, for example, a spring material, and may protrude in an arc shape from the engaging surface **26b**. In addition, as illustrated in FIG. **4B**, the engaging surface **26b** does not have to be a flat surface, and may be a convex curved surface that can be engaged with the tapered surface **16c** of the adapter plate **16**. Other configurations and effects of the transport device and the stamp tool of the present embodiment are the same as those of the first or second embodiment, and a detailed description thereof will be omitted.

(78) Note that the present invention is not limited to the above-described embodiments, and various modifications can be made within the scope of the present invention.

(79) For example, the stamp tool is not limited to the stamp tool **10** of the embodiment described above, and other stamp tools can be used. In addition, the transport head **22** may be provided with at least one of an electrostatic adsorption mechanism, a fitting mechanism, and a screwing mechanism as a secondary attaching means other than the clamping mechanism **26**. By providing these mechanisms to the transport head, it becomes easy to set the mounting force **F3** to the mounting surface **16a** of the adapter plate **16** by the transport head **22** to be larger than the fixing force **F1** of the element **32r** (**32g**, **32b**) to the substrate **30**.

(80) In addition, in the above-described embodiments, as the primary attaching means of the transport head **22**, vacuum adsorption using vacuum suction holes is used. However, in the present invention, it may be unnecessary to use vacuum adsorption, and the stamp tool **10** may be

detachably attached on the transport head **22** only by the secondary attaching means such as the clamping mechanism **26**.

(81) In addition, the transport device **20** according to the present embodiment is used to pick up the elements **32r** (**32g**, **32b**) from the element forming substrate **30**. However, the application is not limited thereto, and the transport device **20** may be used to pick up the elements **32r** (**32g**, **32b**) from a substrate (sheet) having an adhesion layer transferred from the substrate **30** by the laser lift method, etc.

(82) In addition, the transport device **20** according to the present embodiment may be used to pick up an element other than the elements **32r**, **32g**, and **32b** for red, green, and blue light emission. A fluorescent element, etc. may be indicated as another display element. In addition, the other element is not limited to the display element, and may be an electronic element such as a light receiving element, a ceramic capacitor, or a chip inductor, or a semiconductor element.

EXPLANATIONS OF LETTERS OR NUMERALS

(83) **10** STAMP TOOL **11** PROJECTION **12** STAMP LAYER **14** SUPPORT PLATE **14a** INCLINED SURFACE **14b** ADSORBABLE SURFACE **14c** INSERTABLE SURFACE **15** ADHESIVE LAYER **16** ADAPTER PLATE **16a** MOUNTING SURFACE **16b** ADHESIVE SURFACE **16c** TAPERED SURFACE **16d** EDGE PORTION **18** SHIM PLATE **20** TRANSPORT DEVICE **22** TRANSPORT HEAD **24** ADSORBING SURFACE **26** CHUCK MECHANISM **26a** CLAW PORTION **26b** ENGAGING SURFACE **26c** ENGAGING PROJECTION **28** OPENING/CLOSING MECHANISM **30** ELEMENT FORMING SUBSTRATE **32r**, **32g**, **32b** ELEMENT **40** INSTALLATION TABLE **50** FIRST TRANSFER SUBSTRATE (SHEET) **52** ADHESION LAYER **60** SECOND TRANSFER SUBSTRATE (SHEET) **62** ADHESION LAYER **70** MOUNTING SUBSTRATE **80** BASE **82** STAGE **83** GAS FLOW HOLE **84** TOP SURFACE **85** SUCTION HOLE **86** HOUSING RECESS **88** GUIDE MEMBER **89** INCLINED SURFACE **90** POSITIONING MEMBER **92** TIP SURFACE

Claims

1. A stamp tool comprising: a stamp layer having a portion allowing a transport object element to detachably adhere thereto; a support plate, the stamp layer being fixed to the support plate; and an adapter plate replaceably attached to the support plate and having a mounting surface, a transport head being allowed to be detachably attached thereon.
2. The stamp tool according to claim 1, wherein the support plate is replaceably attached to the adapter plate by an adhesive layer.
3. The stamp tool according to claim 1, wherein: the transport object element comprises elements formed on a surface of a substrate; projections corresponding to the elements are formed on the stamp layer; and the elements detachably adhere to the respective projections.
4. The stamp tool according to claim 1, wherein the support plate comprises a glass plate or a ceramic plate having a flat surface.
5. The stamp tool according to claim 1, wherein a tapered surface having an outer diameter decreasing toward the support plate is formed on a side surface of the adapter plate.
6. The stamp tool according to claim 1, wherein a maximum width of the adapter plate along a first axial direction is set to be larger than a width of the support plate.
7. The stamp tool according to claim 5, wherein an insertable surface facing the tapered surface of the adapter plate is provided on a surface of the support plate at a side of the adapter plate.
8. The stamp tool according to claim 1, wherein an adsorbable surface is formed around the stamp layer on a surface of the support plate at a side of the stamp layer.
9. The stamp tool according to claim 1, wherein a shim plate for adjusting a degree of parallelism of the support plate is interposed between the stamp layer and the adapter plate.
10. A method of manufacturing an element array, the method comprising: taking out transport

object elements from a substrate simultaneously and transporting the elements using the stamp tool according to claim 1.

11. A transport device comprising: a stamp tool; and a transport head configured to detachably transport the stamp tool, wherein: the stamp tool includes: a stamp layer allowing a transport object element disposed on a substrate with a predetermined fixing force to adhere into the stamp layer with predetermined adhesion force; and an adapter plate configured to support the stamp layer, and having a mounting surface allowing the transport head to be detachably attached thereon, a mounting force to the mounting surface of the adapter plate by the transport head is greater than the fixing force, the adhesion force of the stamp layer to the transport object element is greater than the fixing force, and a tapered surface having an outer diameter decreasing toward the stamp layer is formed on a side surface of the adapter plate.

12. The transport device according to claim 11, wherein the mounting force to the mounting surface of the adapter plate by the transport head includes a primary mounting force by a primary attaching means and a secondary mounting force by a secondary attaching means other than the primary attaching means.

13. The transport device according to claim 12, wherein the primary attaching means includes a vacuum suction hole formed in the transport head.

14. The transport device according to claim 12, wherein the secondary attaching means includes at least one of a clamping mechanism for detachably attaching the adapter plate to the transport head, an electrostatic adsorption mechanism, a fitting mechanism, and a screwing mechanism.

15. The transport device according to claim 14, wherein a claw portion of a clamping mechanism included in the secondary attaching means is allowed to be engaged with the tapered surface formed on the side surface of the adapter plate.

16. A transport device comprising: a stamp tool; and a transport head configured to detachably transport the stamp tool, wherein: the stamp tool includes: a stamp layer allowing a transport object element disposed on a substrate with a predetermined fixing force to adhere into the stamp layer with predetermined adhesion force; and an adapter plate configured to support the stamp layer, and having a mounting surface allowing the transport head to be detachably attached thereon, a mounting force to the mounting surface of the adapter plate by the transport head is greater than the fixing force, the adhesion force of the stamp layer to the transport object element is greater than the fixing force, the stamp tool further includes a support plate on which the stamp layer is fixed, and the adapter plate is replaceably attached to the support plate.

17. The transport device according to claim 16, wherein the support plate is replaceably attached to the adapter plate by an adhesive layer.

18. The transport device according to claim 11, wherein: the transport object element comprises elements formed on a surface of a substrate; a plurality of projections corresponding to the elements is formed on the stamp layer; and the elements detachably adhere to the respective projections.

19. A method of manufacturing an element array, the method comprising: taking out and transporting a plurality of elements as transport object elements from the substrate simultaneously and transporting the elements using the transport device according to claim 11.
